Karthik Dantu **Curriculum Vitae** **September 2025**

CONTACT

Computer Science and Engineering
University at Buffalo, State University of New York
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RESEARCH INTERESTS

I am interested in real-world applications of robot autonomy. In particular, I am interested in perception challenges in real-world scenarios. Recently, my group has been focused on two specific applications - infrastructure inspection and autonomous excavation.
My research is publicly funded. Correspondingly, all my research work is public. My papers are openly available on my homepage at https://cse.buffalo.edu/faculty/kdantu/ as well as on sites such as Google Scholar. All the code produced by my research group is publicly available on GitHub at https://github.com/droneslab.

EDUCATION

University of Southern California Los Angeles, CA
Department of Computer Science
Ph.D., Computer Science, May 2010

- Dissertation: Reconfigurable Sensor Networks
- Advisor: Prof. Gaurav S. Sukhatme

M.S., Computer Science

- Dissertation: Power-awareness in embedded systems
- Advisor: Prof. Massoud Pedram

Sri Jayachamarajendra College of Engineering Mysore, India
Department of Computer Science and Engineering
B.E., Computer Science and Engineering, 1999

- Dissertation: Implementing SACK and T/TCP in Linux 2.0.36
- Advisor: Prof. Kumar Sivarajan

RESEARCH EXPERIENCE

Associate Professor *Jan 2020—present*
Assistant Professor *Aug 2013—Dec 2019*
SUNY-Buffalo Buffalo, NY
Co-Director, MS Robotics *June 2024—*
Founding Director, Center for Embodied Autonomy and Robotics *Fall 2022—*
Department of Computer Science and Engineering
Distributed Robotics and Networked Embedded Systems (DRONES) Lab
Co-Director, Reliable Mobile Systems (with Prof. Steve Ko and Prof. Lukasz Ziarek)
Postdoctoral Fellow *March 2010—Aug 2013*
Harvard University Cambridge, MA
School of Engineering and Applied Sciences
RoboBees Project
Graduate Researcher *Jan 2001—Jan 2003*
University of Southern California Los Angeles, CA
Research Intern *May 2005 — Aug 2005*
UC Berkeley Earthquake Engineering Research Center Berkeley, CA

TEACHING EXPERIENCE

University at Buffalo Buffalo, NY

Department of Computer Science and Engineering

CSE 468/568: Robotics Algorithms *Sp 14-16,18, 23 Fa 16,18,21-25*

- Completely re-designed course introducing robotics to seniors and first year graduate students - Spring 23 based on F1tenth platform
- Material included kinematics, probabilistic algorithms for localization and mapping, planning, and navigation
- Students also implemented algorithms on Robot Operating System (ROS)
- Course had homeworks, midterm, and finals along with five programming assignments

CSE 610: Learning for Autonomous Systems *Spring 2022*

- Co-taught with Prof. Souma Chowdhury (MAE)
- Study state-of-the-art methods for multi-robot coordination
- Projects using various reinforcement learning approaches to perform such coordination
- Applications include Unmanned Aerial Mobility vertiport coordination, collective construction, soccer gameplay and others

CSE 622: Advanced Computer Systems *Fall 13, 14, 16, 18*

- Co-taught with Steve Ko
- Advanced graduate course dealing with Android internals
- Studied papers on modern computer systems
- Projects based on the Android kernel and framework
- 2013: Resulted in one masters thesis and a paper in EUC 2016
- 2014: Resulted in HotStorage '15 paper, an IoTDI '16 paper and two masters theses
- 2016: Resulted in one masters thesis

CSE 421/521: Operating Systems *Fall 17*

- Concurrency, processes, threads, scheduling
- Memory management
- I/O
- Assignments based on PintOS, an open-source academic OS
- Students implemented scheduler, system calls and virtual memory
- Course also had a midterm and a final

CSE 610: Intelligent Sensing Systems *Spring 17*

- Advanced graduate course on modern sensor systems
- Studied topics related to coordinate frames and visual sensing
- Read papers on sensing algorithms and systems
- Projects based on developing novel applications/algorithms for UAVs/augmented reality
- All projects on a quadrotor platform or the Microsoft Hololens

CSE 749:Multi-Robot Coordination *Fall 15*

- Seminar course on Multi-Robot coordination
- Read papers related to multi-robot systems
- Several research projects related to the field of multi-robot systems

- Wrote embedded Java software for the high end oscilloscopes of Tektronix
- Organized special interest group meetings on Design Patterns in Java

AWARDS

- **2022 IEEE Region 1 Technological Innovation (Academic) Award**, IEEE, Sep 2022.

- **Senior Member**, IEEE 2022.
- **Best poster**, Hotmobile 2022.
- **Best paper**, IEEE Symposium on Multirobot and Multiagent Systems (MRS) 2021.
- **Senior Researcher of the year**, UB CSE 2021.
- **Runner-up**, F1tenth autonomous racing competition held at IROS 2021.
- **NSF CAREER Award**, June 2019.
- **Honorable Mention, Best Paper Award** EICS 2019.
- **Runner-up, best poster award** IPSN 2017
- **Distinguished Research Mentor Award**, University at Buffalo Collegiate Science and Technology Entry Program 2016.
- **Best Demo**, IPSN-SPOTS 2012
- **Runner-up, Best paper award**, IPSN-SPOTS 2012
- Internation Students Assembly, **Outstanding Service Award** 2010
- USC Robotics Lab, **George M. Bekey Outstanding Service Award**, 2010
- NSF **Travel Grant**, SenSys 2005
- Conference **Travel Grant**, SIGCOMM 2002

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE)
- Association for Computing Machinery (ACM)
- USENIX
- AAAS

Research Supervision

Current Postdoctoral Fellows (2)

1. *Charuvahan Adhivarahan* (Summer 2023-)
   Research: Perception and Exploration challenges in multi-robot systems
2. *Young-jin Kim* (Fall 2024-)
   Research: Safe control and planning in mobile robots

Past Ph.D. Students (7)

1. *Sharath Chandrashekhara* (joint with Steve Ko) PhD student (Fall 2014-2018)
   Thesis: Flexible Data Management in Mobile Systems
   Graduated: Fall 2018.
   Employment: Samsung Research, Mountain View, CA.
2. *Zakieh Hashemifar* PhD Student (Fall 2014-Spring 2019)
   Thesis: Visual Simultaneous Localization and Mapping in Indoor Environments
   Graduated: May 2019.
   Employment: Zoox Inc.
   - IPSN '17 Best-poster runner-up
   - Grace Hopper Conference scholarship awardee 2016
   - UB SEAS Dean's fellowship 2014
3. *Farshad Ghanei* PhD Student (Fall 2014-Fall 2020)
   Thesis: Energy Awareness in Mobile Embedded Systems
   Graduated: December 2020.
   Employment: Assistant Teaching Professor, CSE University at Buffalo.

- Graduate Leadership Award, UB CSE 2017
- Best Poster, UB CSE annual research symposium 2017
- UB SEAS Dean's fellowship 2014

4. *Ali Ben Ali* PhD Student (Fall 2015-Spring 2021)
   Thesis: Edge-Assisted, Concurrent Localization and Mapping Systems for Mobile Devices
   Graduated: May 2021.
   Employment: Assistant Teaching Professor, CS Binghamton University.
5. *Chang-Min Park* (joint with Steve Ko) PhD student (Fall 2017–Fall 2022)
   Thesis: Providing Image Confidentiality and Integrity on Mobile Devices
   Graduated: Dec 2022.
   Employment: Software Engineer, Yahoo Inc.
6. *Charuvahan Adhivarahan* PhD student (Fall 2016–Spring 2023)
   Project: Distributed Mapping in Large Urban Environment
   Graduated: May 2023.
   Employment: Postdoctoral Fellow, University at Buffalo.
7. *Yuyang Chen* PhD Student (Fall 2016–Spring 2024)
   Project: Active Perception on Motion-Decoupled Sensor-Robot Platforms
   Graduated: May 2024.
   Employment: Zoox Inc.

Ph.D. Students Supervising - Primary advisor (8), Co-advisor (1)

1. *Sofiya Semenova* PhD student (Fall 2018–)
   Project: Systems support for Edge Vision Applications
   Expected Graduation: Summer 2025.
2. *Timothy Chase* PhD student (Fall 2020–)
   Project: Perception Challenges in Space Autonomy
   Expected Graduation: Summer 2025.
3. *Vaishali Maheshkar* PhD student (Spring 2022–)
   Project: Low-power sensing systems
   Expected Graduation: Fall 2025.
4. *Yash Turkar* PhD student (Spring 2022–)
   Project: LiDAR-based SLAM
   Expected Graduation: 2027.
   - Outstanding leadership award for MS student - 2023.
5. *Kartikeya Singh* PhD student (Fall 2022–)
   Project: Learnable SLAM systems
   Expected Graduation: 2027.
6. *Pranay Meshram* PhD student (Spring 2023–)
   Project: Improving learning-driven perception
   Expected Graduation: 2027.
7. *Shekoufeh Sadeghi* PhD student (Fall 2023–)
   Project: Field-of-view Optimization for Robot Perception
   Expected Graduation: 2028.

8. *Long Duong* PhD student (Fall 2024–)
   Project: Multi-modal sensing for plastics detection
   Expected Graduation: 2029.

   - Dean's fellowship - 2024.

   Co-advisor: Roshan Ayyalasomayajula.
9. *Christo Aluckal* PhD student (Spring 2025–)
   Expected Graduation: 2029.

   - Outstanding service award for MS student - 2025.

Masters Theses (4)

1. *Christo Aluckal* Masters thesis (2024)
   Thesis: Simulation Environment for excavation autonomy.
   Outstanding service award for MS student - 2025
   First job: PhD student.
2. *Praneeth Behara* Masters Thesis (2017-18)
   Thesis: Distributed Robot Mapping
   First job: Intellibot Robotics.
3. *Yash Upadhyay* Masters Thesis (May 2015)
   Thesis: Heart Rate Monitoring and User Behaviour Anomaly Detection Using Dynasense
   Paper: Pratik Lade, Yash Upadhyay, Karthik Dantu, Steven Y. Ko, "Developing Adaptive Quantified-Self Applications Using DynaSense", *2016 IEEE First International Conference on Internet-of-Things Design and Implementation (***IoTDI '16***)*, Berlin, April 2016.
   First Job: Microsoft.
4. *Pratik Lade* Masters Thesis (May 2015)
   Thesis: Calorie Tracking And Sleep Monitoring Using Dynasense
   Paper: Pratik Lade, Yash Upadhyay, Karthik Dantu, Steven Y. Ko, "Developing Adaptive Quantified-Self Applications Using DynaSense", *2016 IEEE First International Conference on Internet-of-Things Design and Implementation (***IoTDI '16***)*, Berlin, April 2016.
   First job: Microsoft.
5. *Varun Anand* *Masters thesis (May 2014)*
   Thesis: Multi-interface Connectivity in Modern Mobile Devices
   Paper: Varun Anand, Karthik Dantu, Steve Ko, Dimitrios Koutsonikolas, "Multi-Interface Connectivity On Modern Mobile Devices", accepted to *14th IEEE/IFIP International Conference on Embedded and Ubiquitous Computing (***EUC '16***)*, August 2016, Paris, France.
   First job: Akamai Technologies, Boston, MA.

Masters Advisees (18)

1. *Umesh Ghaskata* Masters student (2024-)
   Project: UAV coverage path planning.

2. *Yashom Dighe* Masters student (2022-23)
   Project: Differential Flatness based trajectory following on F1tenth.
3. *Smit Rajguru* Masters student (2022-23)
   Project: Slip modeling on F1tenth.
4. *Christo Aluckal* Masters student (2022-24)
   Project: Simulation Environment for Excavation Autonomy.
5. *Ninad* Masters student (2022-)
   Project: AMCL on F1tenth.
6. *Hariprasath Parthasarathy* Masters student (2018-20)
   Project: UB-ANC Emane integration
7. *Arun Suresh* Masters student (2018-20)
   Project: UB-ANC Emane integration
8. *Yash Saraf* Masters student (2018-20)
   Project: Activity recognition on mobile devices
9. *Mrinalini Upadhyaya* Masters student (2018-20)
   Project: Activity recognition on mobile devices
10. *Rakesh Rana* Masters Student (2017-19)
    Project: UAV-based water velocitmetry
11. *Aditya Kohli* Masters Student (2017-19)
    Project: Video Analytics First Job: Google.
12. *Alex Simeonov* Masters Student (Fall 2016, Spring 2017)
    Project: Reptor, Autoclicker
    Graduated: Spring 2017.
13. *Kyle Marcus* Masters Student (Fall 2014, Spring 2015)
    Project: Resource Accounting in Mobile Systems
    Graduated: Spring 2015.
    First Job:
14. *Kaustubh Vartak* Masters Student (Aug 2013–Dec 2014)
    Topic: Kinematics and Gait Design for Daedalus Robot
    First employment: Bloomberg
15. *Mohammadanas Saiyed* Masters Student (Aug 2013–May 2015)
    Topic: Design of Multi-Kinect Tracking System
16. *Manish Jain* *Masters student (May 2014)*
    Topic: Resource Accounting in Mobile Systems
    First job: Alcatel Lucent, Sunnyvale, CA.
17. *Amit Kulkarni* *Masters student (May 2014)*
    Topic: Resource Accounting in Mobile Systems
    First job:RFSpot, Sunnyvale, CA.
18. *Satyaditya Munipalle* *Masters student (May 2014)*
    Topic: Record and replay in mobile devices
    First job: Paypal Inc, Mountainview, CA.

Undergraduate Advisees (9)

1. *Harjot Singh* CE (Fall 2021-Fall 2022)
   Project: Autonomous racing with F1tenth.
2. *Alex Liu* CS (Fall 2018-Spring 2019)
   Project: Application of Machine Learning to sports analysis First Job: Graduate student, U-Michigan.
3. *Tyler Perison* CS (Spring 2018-Fall 2018)
   Project: Data Collection for Visual SLAM
4. *Gitanjali Nandi* CS (Fall 2017-Spring 2018)
   Project: Crazyswarm
5. *Nicholas Ceccarelli* CS (Spring 2018 - Spring 2019)
   Project: Planters
6. *Peter VanNostrand* EE (Spring 2018 - Spring 2019)
   Project: ORB-SLAM on edge devices First job: PhD student, WPI.
7. *Javier Yu* MAE (Fall 2016-Spring 2018)
   Project: UAV Swarms
   First job: PhD student in Robotics at Stanford.
8. *Anand Balakrishnan* CE (Fall 2016-Spring 2018)
   Project: Data Collection for Visual SLAM and WiFi.
   First job: PhD student at USC.
9. *Bob DeBortoli* CS (Fall 2015-Spring 2017)
   Project: MAV sensing
   First Job: PhD student in Robotics at Oregon State.
10. *Manomit Bal* CE (Fall 2015-Spring 2017)
    Project: Multi-camera Localization/Tracking
    First Job: Discovery Robotics.

GRANT SUPPORT

Title:	**Hyperspectral Imaging and Edge Intelligence for Precision Agriculture**
Agency:	NASA
Total:	$400,000
UB Share:	$120,000
UB Credit:	30%
My Credit:	30%
Role:	Co-PI
Effective Dates:	Apr 2025 - June 2026.

Title:	**GRAPPLE: Generalizable & Robust Activity Planning enabled by Physics-based abstraction, simulation and Learning Environments**
Agency:	Office of Naval Research - Science of AI program
Total:	$700,000
UB Share:	$700,000
UB Credit:	100%
My Credit:	25%
Role:	Co-PI
Other PIs/Co-PIs:	Souma Chowdhury, Ehsan Esfahani, Chen Wang
Effective Dates:	July 2023 - June 2026.

Title:	**Simulation Environment to train teachers in Science of Reading**
Agency:	SUNY
Total	$250,000
UB Share:	$250,000
UB Credit:	100%
My Credit:	60%
Role:	PI
Other PIs/Co-PIs:	Srirangaraj Setlur
Effective Dates:	Feb 2025 - Jan 2026.

Title:	**Improving Safety in Robot Autonomy**
Agency:	MOOG Gift
Total:	$750,000
UB Share:	$750,000
UB Credit:	100%
My Credit:	100%
Role:	PI
Other PIs/Co-PIs:	None
Effective Dates:	Apr 2024 - May 2027.

Title:	**Active Perception and Reasoning in Robotics**
Agency:	MOOG Endowment
Total:	$100,000
UB Share:	$100,000
UB Credit:	100%
My Credit:	100%
Role:	PI
Other PIs/Co-PIs:	None
Effective Dates:	Sep 2022 - Aug 2024.

Title:	**Generalizing Robot Perception for Indoor and Outdoor Scenarios by Combining Spatial and Semantic Reasoning**
Agency:	NSF
Total:	$99,563
UB Share:	$99,563
UB Credit:	100%
My Credit:	100%
Role:	PI
Other PIs/Co-PIs:	PB Sujit, IISER, India
Effective Dates:	May 2022 - May 2023.

Title:	**ZTSwarm - Zero Touch Robust Swarm Configuration in the Wild**
Agency:	AFRL
Total:	$1,600,000
UB Share:	$1,600,000
UB Credit:	100%
My Credit:	12.5%
Role:	Co-PI
Other PIs/Co-PIs:	Zhangyu Guan (PI), Nicholas Mastronarde (Co-PI)
Effective Dates:	Jan 2023 - Dec 2025.

Title:	**SSISTL Pilot - SOTER - Safe, Efficient Last Mile Delivery using Autonomous Ground/Air Vehicles**
Agency:	UB - SSISTL
Total:	$58,816
UB Share:	$58,816
UB Credit:	100%
My Credit:	50%
Role:	PI
Other PIs/Co-PIs:	Jinjun Xiong (Co-PI), Chase Murray (Co-PI)
Effective Dates:	Mar 2022 - Feb 2023.

Title:	**AFRL Directional Airborne Networks for Contested Environments (DANCE) - DeepMission Toolchain**
Agency:	AFRL
Total:	$285,235
UB Share:	$285,235
UB Credit:	100%
My Credit:	33%
Role:	Co-PI
Other PIs/Co-PIs:	Zhangyu Guan (PI), Nicholas Mastronarde (Co-PI),
Effective Dates:	Mar 2022 - Feb 2023.

Title:	**DURIP: Reasons: Resilient, Adaptive, Scalable, Autonomy in Networked Swarms**
Agency:	AFOSR
Total:	$471,416
UB Share:	$471,416
UB Credit:	100%
Role:	PI
Other PIs/Co-PIs:	Souma Chowdhury (co-PI), Chunming Qiao (Co-PI), Chase Murray (Co-PI))
Effective Dates:	Feb 2022 - Jan 2024.

Title:	**EFRI E3P: Valorization of Plastic Waste via Advanced Separation and Processing**
Agency:	National Science Foundation
Total:	$1,999,998
UB Share:	$1,999,998
UB Credit:	100%
Role:	Co-PI
Other PIs/Co-PIs:	Paschalis Alexandridis (PI), Marina Tsianou (Co-PI), Javid Rzayev (Co-PI), Luis Velarde (Co-PI)
Effective Dates:	Sep 2020 - Aug 2024.

Title:	**Evolving Neural architectures with Human Augmented Novelty for Complex Environments (ENHANCE)**
Agency:	DARPA
Total:	$465,963
UB Share:	$465,963
UB Credit:	100%
Role:	Co-PI
Other PIs/Co-PIs:	Souma Chowdhury (PI), Ehsan Esfahani (Co-PI), David Doermann (Co-PI)
Effective Dates:	Sep 2019 - Aug 2021.

Title:	**CAREER: Enabling Seamless Vision Sensing in Cloud-Edge Systems**
Agency:	National Science Foundation
Total:	$549,370
UB Share:	$549,370
UB Credit:	100%
Role:	PI
Other PIs/Co-PIs:	None
Effective Dates:	June 2019 - May 2024.

Title:	**Integration of EMANE and the UB-ANC Emulator**
Agency:	Air Force Research Labs
Total:	$50,000
UB Credit:	100%
Role:	Co-PI
Other PIs/Co-PIs:	Nicholas Mastronarde (UB)
Effective Dates:	Jan 2019 - Jan 2020.

Title:	**CRI:CI-NEW:COLLAB: Extensible Software Enabled Unmanned Aerial Vehicles**
Agency:	National Science Foundation
Total:	$900,000
UB Share:	$558,414
UB Credit:	50%
Role:	Co-PI
Other PIs/Co-PIs:	Lukasz Ziarek (UB), David Liu (SUNY-Binghamton)
Effective Dates:	Oct 2018 - Sep 2022.

Title:	**President's Circle Award: UB Robotics Day**
Agency:	University at Buffalo President's office
Total:	$30,000
UB Credit:	100%
Role:	PI
Other PIs/Co-PIs:	Nils Napp (UB), Chunming Qiao (UB)
Effective Dates:	Oct 2017 - Dec 2019.

Title:	**CSR: Small: Project T: Enabling Open Innovation in Mobile Platforms through API Virtualization**
Agency:	National Science Foundation
Total:	$500,000
UB Credit:	100%
Role:	Co-PI
Other PIs/Co-PIs:	Steve Ko (UB)
Effective Dates:	Aug 2016-July 2019.

Title:	**Community of Excellence on Sustainable Manufacturing and Advanced Robotics Technologies (SMART CoE)**
Agency:	University at Buffalo
Total:	$4,700,000
UB Credit:	100%
Role:	Co-PI
Other PIs/Co-PIs:	Kemper Lewis, Omar Khan and others
Effective Dates:	Aug 2015-July 2020.

Title:	**RI: Medium: Collaborative Proposal: Novel Depth Sensor Design and Sensing Algorithms for Flapping-Wing Micro-Aerial Vehicles**
Agency:	National Science Foundation
Total:	$1,088,697
UB Share:	$372,655
UB Credit:	33%
Role:	Lead PI
Other PIs/Co-PIs:	Rob Wood (Harvard), Sanjeev Koppal and Huikai Xie (U-Florida)
UB Credit:	35%
Effective Dates:	Aug 2015-July 2019.

Supplements and Gifts

Title:	**REU Supplement: Dataset collection to identify patterns of interaction across mobile devices**
Agency:	National Science Foundation
Total:	$14,000
Role:	PI
Other PIs/Co-PIs:	None
Effective Dates:	May 2018-July 2019.

Title: **ROA Supplement: Incorporating WiFi Sensing in RGB-D Mapping**
Agency: National Science Foundation
Total: $30,000
UB Credit: 100%
Role: PI
Other PIs/Co-PIs: Deborah Burhans (Canisius College)
Effective Dates: May 2017-July 2019.

Title: **REU Supplement: Imitating RoboBee dynamics using small quadrotors**
Agency: National Science Foundation
Total: $8,000
Role: PI
Other PIs/Co-PIs: None
Effective Dates: May 2016-July 2019.

Title: **Intel Galileo Donation Program**
Company: Intel Corp.
Role: PI
Date Received: 1/2014
UB Amount: 10 Intel Galileo development boards

SERVICE

Government Grant Agencies

- Reviewer, NSF Panel - 2015, 2016, 2017, 2019, 2022, 2023, 2024, 2025.
- Reviewer, Research Grants Council of Hong Kong - 2015, 2016.
- Reviewer, National Research Fund, Luxembourg - 2023.

Research Program Committees

- 2026: Sensys
- 2025: Sensys, ICRA (reviewer), IROS (reviewer)
- 2024: ICRA (reviewer), IROS (reviewer)
- 2023: **PC:** Mobicom, MobiSys, Sensys, ICRA (reviewer), IROS (reviewer)
- 2022: **PC:** Mobicom, DARS
- 2021: **PC:** MobiSys, ICDCS, MRS
- 2020: **PC:** Mobicom, DARS
- 2019: **PC:** Mobicom, IWQoS, MASS, MRS
- 2018: **Co-organizer:** First workshop on Internet of People, Assistive Robots, and Things (IoPARTS) with MobiSys 2018.
  PC: ACM Sensys, DARS, IEEE MASS.
  Publicity Chair: COMSNETS 2019.
  Associate Editor: ICRA.
- 2017: **Co-organizer:** RSS workshop on "Multi-Robot Communication In The Wild"
  PC: SECON, ACM Sensys, ICCCN
  Publicity Chair: SECON
- 2016: **PC:** ACM Sensys, IPSN, DARS, COMSNETS
  Associate Editor: ICRA
  Poster Chair: Sensys
- 2015: **PC:** IPSN, ROBIO
  Co-organizer: 2nd Workshop on Robotic Sensor Networks during CPSWeek

- 2014: **PC:** DARS, EWSN, AAAI
  Associate Editor: ICRA,
  Poster/Demo Chair: DCOSS
  Organizer: Northeast Robotics Colloquium 2014
  Co-organizer: Workshop on Robotic Sensor Networks during CPSWeek
- 2013: **PC:** S-CUBE, Sensornets
- 2012: **PC:** Sensornets
  Publication Chair: Sensys

REVIEWING

Journals - frequent

Springer Autonomous Robots
IEEE Robotics and Automation Letters
IEEE Transactions on Mobile Computing

Journals - Occasionally

ACM Transactions on Sensor Networks
Journal of Sensors
IEEE Transactions on Automation Science and Engineering (T-ASE)
IEEE Transactions on Parallel and Distributed Systems (TPDS)
IEEE Transactions on Wireless Communications (TWC)
IEEE Journal on Selected Areas in Communications
ACM Transactions on Sensor Networks (TOSN).

Conferences

Robotics: Science and Systems
IEEE International Conference on Robotics and Automation (ICRA)
IEEE/RSJ International Conference on Robots and Systems (IROS)
Workshop on the Algorithmic Foundations of Robotics (WAFR)
IEEE Workshop on Embedded Network Sensors (Emnets)
IEEE/ACM International Conference on Information Processing in Sensor Networks (IPSN)
IEEE International Conference in Distributed Computing in Sensor Systems (DCOSS)
IEEE International Conference on Design Automation and Testing in Europe (DATE)
ACM/IEEE International Design Automation Conference (DAC)
IEEE Conference on Sensors, Mesh and Adhoc Networking and Communications (SECON)
IEEE Conference on Automation Science and Engineering (CASE)

CONFERENCE / JOURNAL (H-INDEX=20, CITATIONS=2533)

Accepted/Published: Journals

1. Sofiya Semenova, Steven Ko, Yu David Liu, Lukasz Ziarek, and Karthik Dantu. A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems. ACM Trans. Embed. Comput. Syst. 24, 1, Article 2 (January 2025), 31 pages.
2. Chen, Yuyang, Dingkang Wang, Lenworth Thomas, Karthik Dantu, and Sanjeev J. Koppal. ”Design of an adaptive lightweight LiDAR to decouple robot-camera geometry.” IEEE Transactions on Robotics (2024).

3. Stavinski, N., Maheshkar, V., Thomas, S., Dantu, K. and Velarde, L.. Mid-infrared spectroscopy and machine learning for postconsumer plastics recycling. Environmental Science: Advances, 2(8), pp.1099-1109, July 2023.
(**Best paper runner-up for 2023**)
4. Ali J. Ben Ali, Marziye Kouroshli, Sofiya Semenova, Zakieh Sadat Hashemifar, Steven Y. Ko, and Karthik Dantu. "Edge-SLAM: Edge-Assisted Visual Simultaneous Localization and Mapping.", ACM Transactions on Embedded Computing Systems (TECS) 22, 1, Article 18 (January 2023), 31 pages.
5. Zhengxiong Li, Baicheng Chen, Xingyu Chen, Chenhan Xu, Yuyang Chen, Feng Lin, Changzhi Li, Karthik Dantu, Kui Ren, and Wenyao Xu, "Reliable Digital Forensics in the Air: Exploring An RF-based Drone Identification System", accepted to *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (IMWUT) 2022.
6. Yaoli Zhao, Patatri Chakraborty, Nicholas Stavinski, Luis Velarde, Vaishali Maheshkar, Karthik Dantu, Arindam Phani, Seonghwan Kim, Thomas Thundat, "Standoff and Point Detection of Thin Polymer Layers Using Microcantilever Photothermal Spectroscopy", in *Journal of the Electrochemical Society* Focus Issue on Biosensors and Nanoscale Measurements, Volume 169, Number 3, 37501, March 2022.
7. Zaid Tasneem, Charuvahan Adhivarahan*, Dinkang Wang, Huikai Xie, Karthik Dantu, Sanjeev Koppal, "Adaptive Fovea for Scanning Depth Sensors", The International Journal of Robotics Research (IJRR), 39 (7), pp 837–855, July 2020.
8. Zakieh Hashemifar*, Charuvahan Adhivarahan*, Anand Balakrishnan*, Karthik Dantu, "Improving Indoor Visual Localization/Mapping with WiFi Sensing", Springer Autonomous Robots (AuRo) (2019), Volume 43, number 8, pp 2245–2260, Dec 2019.
9. Farshad Ghanei*, Pranav Tipnis*, Kyle Marcus*, **Karthik Dantu**, Steven Y. Ko, Lukasz Ziarek, "OS-Based Resource Accounting for Asynchronous Resource Use In Mobile Systems", accepted to *IEEE Internet of Things Journal (IEEE IoTJ)*, vol. 6, no. 3, pp. 5841-5852, June 2019.
10. Chang Min Park*, Taeyeon Ki*, Ali Ben Ali*, **Karthik Dantu**, Steve Ko, "Mu: Mapping UI Events to Gesture and Voice", in *Proceedings of the ACM on Human-Computer Interaction (PACMHCI)* 3, EICS, Article 5 (June 2019), 22 pages.
11. Jalil Modares*, Nicholas Mastronarde, **Karthik Dantu**, "UB-ANC Emulator: An Emulation Framework for Multi-Agent Drone Networks", in *International Journal of Micro-Air Vehicles (IJMAV)*, April 2019 (13 pages).
12. Yin Yan*, Girish Gokul*, **Karthik Dantu**, Steven Y. Ko, Jan Vitek, Lukasz Ziarek, "Can Android Run on Time? Extending and Measuring the Android Platform's Timeliness", *ACM Transactions on Embedded Computing Systems (ACM TECS)*, Volume 17 issue 6, January 2019 (26 pages).
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Accepted/Published: Conferences

1. Prithvi Poddar, Ehsan Tarkesh Esfahani, Karthik Dantu, and Souma Chowdhury. "Automated Generation of Diverse Courses of Actions for Multi-Agent Operations using Binary Optimization and Graph Learning." accepted to IEEE International Conference on Automation Science and Engineering (CASE), August 2025, Los Angeles, CA.
2. Yash Turkar, Youngjin Kim, and Karthik Dantu. "Active Illumination Control in Low-Light Environments using NightHawk." accepted to 19th International Symposium on Experimental Robotics (ISER), July 2025, Santa Fe, NM.
3. Kristian Dalland, Chen Wang, Karthik Dantu, Souma Chowdhury, Ehsan Esfahani, "Hybrid Human Model for Time of Hike Prediction", accepted to 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), August 2025, Copenhagen, Denmark.
4. Christo Aluckal, Roopesh Vinodh Kumar Lal, Sean Courtney, Yash Turkar, Yashom Dighe, Youngjin Kim, Jake Gemerek, Karthik Dantu, "TERA: A Simulation Environment for Terrain Excavation Robot Autonomy," 2025 IEEE International Conference on Simulation, Modeling, and Programming for Autonomous Robots (SIMPAR), Palermo, Italy, 2025, pp. 1-6, doi: 10.1109/SIMPAR62925.2025.10979147.
5. Timothy Chase, Karthik Dantu, "MARs: Multi-view Attention Regularizations for Patch-Based Feature Recognition of Space Terrain. In Proceedings of the European Conference on Computer Vision (ECCV), Sep 2024.
6. Yuyang Chen, Praveen Masilamani, Bhavin Jawade, Srirangaraj Setlur and Karthik Dantu, "DIOR: Dataset for Indoor-Outdoor Reidentification Long Range 3D/2D Skeleton Gait Collection Pipeline, Semi-Automated Gait Keypoint Labeling and Baseline Evaluation Methods," 2024 IEEE International Joint Conference on Biometrics (IJCB), Buffalo, NY, USA, 2024,
7. Kaleem Nawaz Khan, Ali Khalid, Yash Turkar, Karthik Dantu, and Fawad Ahmad. 2024. VRF: Vehicle Road-side Point Cloud Fusion. In Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services (MOBISYS). Association for Computing Machinery, New York, NY, USA, 547?560.
8. D. Kumar, S. Gopinath, K. Dantu and S. Y. Ko, "JacobiGPU: GPU-Accelerated Numerical Differentiation for Loop Closure in Visual SLAM," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 1687-1693.
9. Xia, Stephen, Minghui Zhao, Charuvahan Adhivarahan, Kaiyuan Hou, Yuyang Chen, Jingping Nie, Eugene Wu, Karthik Dantu, and Xiaofan Jiang. "Anemoi: A low-cost sensorless indoor drone system for automatic mapping of 3d airflow fields." In Proceedings of the 29th Annual International Conference on Mobile Computing and Networking (MOBICOM), pp. 1-16. 2023.
10. Shishir Gopinath, Karthik Dantu, Steve Ko, "Improving the Performance of Local Bundle Adjustment with Efficient Use of GPU Resources", accepted to *International Conference on Robotics and Automation* (ICRA), June 2023, London, UK.
11. Prajit Kumar, Jhoel Witter, Steve Paul, Karthik Dantu, Souma Chowdhury, "Graph Learning based Decision Support for Multi-Aircraft Take-Off and Landing at Urban Air Mobility Vertiports", accepted to *AIAA SCITECH 2023 Forum*, Jan 2023.

12. Chase, Timothy, Ali J. Ben Ali, Steven Y. Ko, and Karthik Dantu. "PRE-SLAM: Persistence Reasoning in Edge-assisted Visual SLAM." In 2022 *IEEE 19th International Conference on Mobile Ad Hoc and Smart Systems* (MASS), pp. 458-466. IEEE, 2022.
13. Timothy Chase Jr, Chris Gnam, John Crassidis, **Karthik Dantu**, "You Only Crash Once: Improved Object Detection for Real-Time, Sim-to-Real Hazardous Terrain Detection and Classification for Autonomous Planetary Landings", accepted to *AAS/AIAA Astrodynamics Specialist Conference*, August 2022, Charlotte NC.
14. Zhengxiong Li, Baicheng Chen, Xingyu Chen, Chenhan Xu, Yuyang Chen, Feng Lin, Changzhi Li, **Karthik Dantu**, Kui Ren, and Wenyao Xu, "Reliable Digital Forensics in the Air: Exploring An RF-based Drone Identification System", accepted to *In International Joint Conference on Pervasive and Ubiquitous Computing* (UbiComp) 2022.
15. Sofiya Semenova, Steven Y. Ko, Yu David Liu, Lukasz Ziarek, **Karthik Dantu**, "A quantitative analysis of system bottlenecks in visual SLAM", In *Proceedings of the 23rd Annual International Workshop on Mobile Computing Systems and Applications* (HotMobile '22). Association for Computing Machinery, New York, NY, USA, pp 74–80. (**Best Poster**)
16. Chris Gnam, Timothy B. Chase, **Karthik Dantu**, John L. Crassidis. "Efficient Feature Matching and Mapping for Terrain Relative Navigation Using Hypothesis Gating," AIAA 2022-2513, *AIAA SCITECH 2022 Forum*, January 2022.
17. Behjat, Amir, Hemanth Manjunatha, Prajit Krisshna Kumar, Apurv Jani, Leighton Collins, Payam Ghassemi, Joseph Distefano, David Doermann, **Karthik Dantu**, Ehsan Esfahani, Souma Chowdhury, "Learning Robot Swarm Tactics over Complex Adversarial Environments." In 2021 *International Symposium on Multi-Robot and Multi-Agent Systems* (MRS), pp. 83-91. IEEE, 2021. (**Best paper**)
18. J. P. Distefano, H. Manjunatha, S. Chowdhury, K. Dantu, D. Doermann and E. T. Esfahani, "Using Physiological Information to Classify Task Difficulty in Human-Swarm Interaction," *2021 IEEE International Conference on Systems, Man, and Cybernetics* (SMC), 2021, pp. 1198-1203.
19. Adam Czerniejewski, John Henry Burns, Farshad Ghanei, Karthik Dantu, Yu David Liu, and Lukasz Ziarek, "JCopter: Reliable UAV Software Through Managed Languages", 2021 *IEEE/RSJ conference on Intelligent Robots and Systems* (IROS), 2021, Prague, Czech Republic, 2021.
20. Xiaozhou Liang, John Burns, Joseph Sanchez, Karthik Dantu, Lukasz Ziarek and David Liu, "Understanding Bounding Functions in Safety-Critical UAV Software," 2021 *IEEE/ACM 43rd International Conference on Software Engineering* (ICSE), 2021, pp. 1311-1322.
21. Leighton Collins, Payam Ghassemi, Ehsan Esfahani, David Doermann, Karthik Dantu, Souma Chowdhury, "Scalable Coverage Path Planning of Multi-Robot Teams for Monitoring Non-Convex Areas," IEEE International Conference on Robotics and Automation (ICRA), June 2021, pp. .
22. Chang Min Park, Donghwi Kim, Deepesh Veersen Sidhwani, Andrew Fuchs, Arnob Paul, Sung-Ju Lee, Karthik Dantu, and Steven Y. Ko. "Rushmore: securely displaying static and animated images using TrustZone", In Proceedings of the 19th Annual International Conference on Mobile Systems, Applications, and Services (MOBISYS). Association for Computing Machinery, New York, NY, USA, 122-135.

23. Hemanth Manjunatha, Joseph Distefano, Apurv Jain, Payam Ghassemi, Souma Chowdhury, David Doermann, Karthik Dantu, Ehsan Esfahani, "Using Physiological Measurements to Analyze the Tactical Decisions in Human Swarm Teams," 2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2020, pp. 256-261, doi: 10.1109/SMC42975.2020.9282845.

24. Ali J. Ben Ali*, Zakieh Sadat Hashemifar*, and Karthik Dantu, "Edge-SLAM: edge-assisted visual simultaneous localization and mapping.", In Proceedings of the 18th International Conference on Mobile Systems, Applications, and Services (MOBISYS). Association for Computing Machinery, New York, NY, USA, 325?337.

25. Zakieh Hashemifar, Karthik Dantu, "Practical Persistence Reasoning in Visual SLAM," IEEE International Conference on Robotics and Automation (ICRA), 2020, pp. 7307-7313.

26. Matthew Rantanen*, Nicholas Mastronarde, Jeff Hudack, Karthik Dantu, " Decentralized Task Allocation in Lossy Networks: A Simulation Study," *IEEE International Conference on Sensing, Communication and Networking (SECON)*, June 2019. (28.6%)

27. Charuvahan Adhivarahan*, Karthik Dantu, "Wisdom: Wireless Sensor Driven Distributed Online Mapping", accepted for publication in *Proceedings of International Conference on Robotics and Automation (ICRA)*, Montreal, Canada, May 2019 (44%).

28. Taeyeon Ki*, Chang Min Park*, Karthik Dantu, Steve Y. Ko, Lukasz Ziarek, "Mimic: Automated UI Testing System for Android Apps", accepted to *The 41st ACM/IEEE International Conference on Software Engineering (ICSE)*, Montreal, Canada, May 2019 (21%).

29. Chang Min Park*, Taeyeon Ki*, Ali Ben Ali*, Karthik Dantu, Steve Ko, "Mu: Mapping UI Events to Gesture and Voice", accepted to *The 11th ACM SIGCHI Symposium on Engineering Interactive Computing Systems (EICS)*, Valencia, Spain, June 2019 (30%) **(Honorable Mention, Best Paper Award)** .

30. Adam Czerniejewski*, Karthik Dantu and Lukasz Ziarek, "jUAV: a Real-time Java UAV Autopilot", In *The Second IEEE International Conference on Robotic Computing (IRC)*, Laguna Hills, CA.

31. Sharath Chandrasekhara*, Taeyeon Ki*, Kyungho Jeon*, Karthik Dantu, Steven Y. Ko, Lukasz Ziarek, "BlueMountain: An Architecture to Customize Data Management on Mobile Systems", To appear in *The 23rd Annual International Conference on Mobile Computing and Networking (MOBICOM)*, Oct 2017, Salt Lake City, UT, USA. (19% acceptance)

32. Yuyang Chen*, Sawyer Fuller, **Karthik Dantu**, "Quadrobee: Simulating Flapping Wing Aerial Vehicle Dynamics On A Quadrotor", In *Proceedings of International Conference on Intelligent Robotics and Systems (IROS)*, Vancouver, September 2017 (45%).

33. Taeyeon Ki*, Alexander Simeonov*, Bhavika Pravin Jain*, Chang Min Park*, Keshav Sharma*, Karthik Dantu, Steven Y. Ko, Lukasz Ziarek, "Reptor: Enabling API Virtualization on Android for Platform Openness", To appear in *The 15th ACM International Conference on Mobile Systems, Applications, and Services (MOBISYS)*, June 2017, Niagara Falls, NY, USA. (18% acceptance)

34. Yin Yan*, Karthik Dantu, Steve Ko, Jan Vitek, Lukasz Ziarek, "Making Android Run on Time", In Proceedings of *2017 IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*, April 2017, Pittsburgh, PA, USA. (21% acceptance)
35. Jalil Modares*, Nicholas Mastronarde, **Karthik Dantu**, "UB-ANC Planner: Energy Efficient Coverage Path Planning with Multiple Drones", accepted for publication in *Proceedings of International Conference on Robotics and Automation (ICRA)*, Singapore, 2017. (41% acceptance)
36. Debra Burhans, Karthik Dantu, "ARTY: Fueling Creativity through Art, Robotics and Technology for Youth", In Proceedings of *The Seventh Symposium on Educational Advances in Artificial Intelligence* (EAAI), Feb 2017, San Francisco, CA.
37. Jalil Modares*, Nicholas Mastronarde, **Karthik Dantu**, "UB-ANC Emulator: An Emulation Framework for Multi-Agent Drone Networks", In *Proceedings of I IEEE International Conference on Simulation, Modeling, and Programming for Autonomous Robots (SIMPAR)*, San Francisco, December 2016.
38. Varun Anand*, Karthik Dantu, Steve Ko, Dimitrios Koutsonikolas, "Multi-Interface Connectivity On Modern Mobile Devices", accepted to *14th IEEE/IFIP International Conference on Embedded and Ubiquitous Computing (EUC)*, August 2016, Paris, France.
39. Farshad Ghanei*, Pranav Tipnis*, Kyle Marcus*, Karthik Dantu, Steve Ko, Lukasz Ziarek, "OS-based Resource Accounting for Asynchronous Resource Use in Mobile Systems", accepted to *IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, August 2016, San Francisco, CA.
40. Pratik Lade*, Yash Upadhyay*, Karthik Dantu, Steve Ko, "Developing Adaptive Quantified-Self Applications Using DynaSense", *IEEE First International Conference on Internet-of-Things Design and Implementation (IoTDI)*, April 2016, Berlin, Germany.
41. Richard Moore, Karthik Dantu, Radhika Nagpal, Geoff Barrows, "Autonomous MAV guidance with a lightweight omnidirectional vision sensor", *IEEE International Conference on Robotics and Automation (ICRA)*, June 2014, Hong Kong, China.
42. Karthik Dantu, Spring Berman, Bryan Kate, Radhika Nagpal, "A Comparison of Deterministic and Stochastic Approaches to Allocating Spatially Dependent Tasks in Micro-Aerial Vehicle Swarms", *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Octobe 2012, Villamoura, Portugal.
43. Bryan Kate, Jason Waterman, Karthik Dantu, Matt Welsh, "Simbeeotic: A Simulator and Testbed for Micro-Aerial Vehicle Experiments", *In 11th International Conference on Information Processing in Sensor Networks (SPOTS Track) (IPSN-SPOTS)*, April 2012, Beijing, China.
Runner-up for best paper award
44. Karthik Dantu, Bryan Kate, Jason Waterman, Peter Bailis, Matt Welsh, "Programming Micro-Aerial Vehicle Swarms with Karma", In 9th International Conference on Embedded Networked Sensor Systems (Sensys '11), November 2011, Seattle, WA.
45. Mi Zhang, Anand Joshi, Ritesh Kadmawala, Karthik Dantu, Sameera Poduri and Gaurav S. Sukhatme, "OCRdroid: A Framework to Digitize Text using Mobile Phones", In ICST International Conference on Mobile Computing, Applications and Services (MobiCASE), October 2009, San Diego, USA.

46. Karthik Dantu, Prakhar Goyal and Gaurav S. Sukhatme, “Relative Bearing Estimation from Commodity Radios”, In IEEE International Conference on Robotics and Automation (ICRA), pp 3871–3877, May 2009, Kobe, Japan.
47. Karthik Dantu, Gaurav S. Sukhatme, “Connectivity vs. Control: Using Directional and Positional Cues to Stabilize Routing in Robot Networks”, In Proceedings of International Conference on Robot Communication and Coordination (ROBOCOMM), pp 1–7, March 2009, Odense, Denmark.
48. Karthik Dantu and Gaurav S. Sukhatme, “Detecting and Tracking Level Sets of Scalar Fields using a Robotic Sensor Network”, In IEEE International Conference on Robotics and Automation (ICRA), pp 3665–3672, April 2007, Rome, Italy.
49. Krishna Chintalapudi, Jeongyeup Paek, Omprakash Gnawali, Tat Fu, Karthik Dantu, John Caffrey, Ramesh Govindan, Erik Johnson, “Structural Damage Detection and Localization Using NetSHM”, In Proceedings of Fifth International Conference on Information Processing in Sensor Networks (IPSN-SPOTS), pp 475–482, April 2006, Nashville, TN.
50. Karthik Dantu, Mohammad H. Rahimi, Hardik Shah, Sandeep Babel, Amit Dhariwal and Gaurav S. Sukhatme, “Robomote: Enabling mobility in sensor networks”, In Fourth IEEE/ACM International Conference on Information Processing in Sensor Networks (IPSN-SPOTS), pp 404–409, April 2005, Los Angeles, USA.
51. Morteza Maleki, Karthik Dantu and Massoud Pedram, “Lifetime Prediction Routing in Mobile Adhoc Networks”, In International Wireless Communications and Networking Conference (WCNC), pp 1185–1190, March 2003.
52. Kihwan Choi, Karthik Dantu, Wei-Chung Cheng and Massoud Pedram, “Frame-based Dynamic Voltage and Frequency Scaling for a MPEG Decoder”, In IEEE/ACM International Conference on Computer-Aided Design (ICCAD), pp 732–737, September 2002, San Jose, USA.
53. Morteza Maleki, Karthik Dantu and Massoud Pedram, “Power-aware Source Routing Protocol for Mobile Adhoc Networks”, In International Symposium on Low Power Electronics and Design (ISLPED), pp 72–75, May 2002, Monterey, USA.

WORKSHOP PUBLICATIONS

1. Sharath Chandrashekhara, Taeyeon Ki, Karthik Dantu, Steve Ko, ”Duvel: Enabling Context-driven, Multi-profile Apps on Android through Storage Sand-boxing”, in *1st International Workshop on Edge Systems, Analytics and Networking (EdgeSys 2018)*, Munich, Germany, pp 31-36.
2. Zakieh Hashemifar, Kyung Won Lee, Karthik Dantu, Nils Napp, ”Geometric Mapping for Sustained Indoor Autonomy”, in *1st International Workshop on Internet of People, Assistive Robots and ThingS (IoPARTS ’18)*, Munich, Germany, June 2018.
3. Matthew Rantanen, Jalil Modares, Nicholas Mastronarde, Farshad Ghanei, Karthik Dantu, ”Performance of the Asynchronous Consensus Based Bundle Algorithm in Lossy Network Environments”, in *10th IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM ’18).*
4. Neeti Narayan, Nishant Sankaran, Devansh Arpit, Karthik Dantu, Srirangaraj Setlur, Venu Govindaraju, ”Person Re-identification for Improved Multi-person Multi-camera Tracking by Continuous Entity Association”, in *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2017 (33% acceptance)

5. Jalil Modares, Nicholas Mastronarde, Karthik Dantu, ”Realistic network simulation in the UB-ANC aerial vehicle network emulator”, In *The 1st IEEE International Workshop on Wireless Communications and Networking in Extreme Environments (WCNEE ’17)*, Atlanta, GA.
6. Adam Czerniejewski, Shaun Cosgrove, Yin Yan, Karthik Dantu, Steven Ko and Lukasz Ziarek, ”jUAV: A Java Based System for Unmanned Aerial Vehicles”, In *The 14th International Workshop on Java Technologies for Real-Time and Embedded Systems (JTRES 2016)*, August 2016, Lugano, Switzerland. (30% acceptance)
7. Girish Gokul, Yin Yan, Karthik Dantu, Steven Ko, Lukasz Ziarek, ”Real Time Sound Processing on Android”, In *The 14th International Workshop on Java Technologies for Real-Time and Embedded Systems (JTRES 2016)*, August 2016, Lugano, Switzerland. (30% acceptance)
8. Kyungho Jeon, Sharath Chandrashekhara, Karthik Dantu, Steven Y Ko, ”Pixelsior: Photo Management as a Platform Service for Mobile Apps”, in *8th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 16)*, June 2016, Denver, CO. (25% acceptance)
9. Yin Yan, Chunyu Chen, Karthik Dantu, Steven Y Ko, Lukasz Ziarek, ”Using a Multi-Tasking VM for Mobile Applications”, In *Proceedings of the 17th International Workshop on Mobile Computing Systems and Applications (HotMobile ’16)*, February 2016, St. Augustine, FL. (32% acceptance)
10. Sharath Chandrashekhara, Kyle Marcus, Rakesh G. M. Subramanya, Hrishikesh S. Karve, Karthik Dantu, and Steven Y. Ko, “Enabling Automated, Rich, and Versatile Data Management for Android Apps with BlueMountain”, *To appear in 7th USENIX workshop on Hot Topics in Storage and File Systems (HotStorage ’15)*, July 2015, Santa Clara, CA (30% acceptance)
11. Karthik Dantu, Bryan Kate, Jason Waterman, Peter Bailis, Matt Welsh, “Coordinating Robotic Bee Swarms”, In NSF Workshop on Pervasive Computing at Scale (PeCS ’11), January 2011, Seattle, USA.
12. Avinash Parnandi, Ken Le, Pradeep Vaghela, Aalaya Kolli, Karthik Dantu, Sameera Poduri and Gaurav S. Sukhatme, “Coarse In-Building Localization with Smartphones”, In ICST International Workshop on Innovative Mobile User Interactivity (IMUI ’09) alongwith MobiCASE 2009, October 2009, San Diego, USA.
13. Dheeraj Kota, Neha Laumas, Urmila Shinde, Saurabh Sonalkar, Karthik Dantu, Sameera Poduri and Gaurav S. Sukhatme, “deSCribe: A Personalized Tour Guide and Navigational Assistant”, In ICST International Workshop on Innovative Mobile User Interactivity (IMUI ’09) alongwith MobiCASE 2009, October 2009, San Diego, USA.
14. Jesse Butterfield, Karthik Dantu, Brian P. Gerkey, Odest C. Jenkins and Gaurav S. Sukhatme, “Autonomous biconnected networks of mobile robots”, In IEEE Workshop on Wireless Multihop Communications in Networked Robotics (WMCNR ’08), pp 640–646, April 2008, Germany.
15. Karthik Dantu and Gaurav S. Sukhatme, “Rethinking data-fusion based services in sensor networks”, In Third Workshop on Embedded Networked Sensors (Emnets ’06), pp 1–6, April 2006, Boston, USA.
16. John Caffrey, Ramesh Govindan, Erik Johnson, Bhaskar Krishnamachari, Sami Masri, Gaurav S. Sukhatme, Krishnakanth K. Chintalapudi, Karthik Dantu, Sumit Rangwala,

Avinash Sridharan, Ning Xu and Marco Zuniga, "Networked Sensing for Structural Health Monitoring", In Fourth International Workshop on Structural Control (IWSC '04), June 2004, New York, USA.

INVITED TALKS

- *Long-term Mapping Challenges in Indoor environments*
  Keynote, 33rd Joint Conference on Communications and Information, Korea — Apr 23.
  IEEE AESS UAS Symposium — Sep 22.
  U-Nevada Reno — Sep 22.
- *Multirobot Perception and Coordination*
  MOOG — Apr 22.
  Air Force Research Labs — Feb 21.
  Army Research Labs — Mar 21.
  Indian Institute of Science — Mar 21.
- *Improving Visual SLAM for Urban Environments*
  Bose Research — June '19.
  Tufts University — Apr '19.
  Mobicom 2019 PC Mini-Research Symposium — May '19.
- *Musings About Research Process, Research Directions and Research Tools in Mobile Systems*
  Keynote, Mobisys 2018 PhD Forum — May '18.
- *Planning and Coordination in Multi-Robot Systems*
  IBM TJ Watson Research — Nov '17.
  TCS Research, India — Jan '18.
- *Coordinating Micro-Aerial Swarms*
  INSIGHTS Program - UB Hayes Society — May '15.
  SRC Corporation — Aug '15.
- *Improving OS and Networking Support for Modern Mobile Devices*
  Microsoft Research India — June '14.
  Adobe Research India — June '14.